Mathematical Formulation of the Proactive Golden-Hour Bridge Model (PGBM) for Post-Earthquake UAV-Assisted Response

Author Name
Department of Computer Science and Engineering
University Name
Email: author@email.com

***Abstract*—This document presents the formal mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM), a UAV-assisted emergency response system for post-earthquake scenarios. The system operates as an online decision-making framework where Responder UAVs are dynamically assigned to detected survivors to provide immediate assistance during the critical "Golden Hour" before ground rescue teams physically arrive. The model captures the coupling between survivor task states, UAV dynamics, battery constraints, time-dependent urgency decay, and 3D routing feasibility. The global system state is formally defined, and the event-triggered decision policy is mathematically characterized.**

I. Introduction

In the aftermath of a major earthquake, Scout UAVs can rapidly detect trapped survivors using thermal and optical sensors. However, Ground Rescue Teams face significant delays due to collapsed infrastructure and blocked roads, creating a deadly gap between detection and physical rescue. The Proactive Golden-Hour Bridge Model (PGBM) introduces an intermediate UAV-assisted response layer where Responder UAVs are dispatched immediately to provide emergency supplies, two-way communication, and GPS relay before ground teams arrive.

A. System Overview

The PGBM consists of four primary actors:

1) **Scout UAVs**: Detect survivors and transmit detection packets.
2) **Command Center**: Maintains global state and executes online allocation/routing decisions.
3) **Responder UAVs**: Execute bridge missions to deliver aid.
4) **Ground Rescue Team**: Performs final physical extraction (outside core model).

B. Core Assumptions

- Obstacles are static during a single mission (Version 1).
- Communication between all actors is reliable.
- Each Responder UAV serves one survivor at a time.
- Survivor tasks are independent and non-preemptive (baseline).

II. Mathematical Notation

Throughout this formulation, bold uppercase letters denote sets, bold lowercase denote vectors, and scalar values use standard notation. The time variable t denotes the current decision time (continuous). All decisions are event-triggered.

TABLE I
Notation Summary

Symbol	Description
$\mathcal{S}(t), \mathcal{D}(t), \mathcal{E}$	Survivor set, Responder UAV set, Environment
$\mathcal{Q}_{\text{high}}, \mathcal{Q}_{\text{low}}$	High/low confidence queues ($c_i \geq c_{\text{thresh}}$)
$S_i = (p_i, t_i^{\text{det}}, c_i, \sigma_i, u_i, a_i)$	Survivor task
$D_j = (p_j, p_j^{\text{base}}, B_j, \text{status}_j)$	Responder UAV
$p_i, p_j, p_j^{\text{base}} \in \mathbb{R}^3$	Positions and base station
$\sigma_i \in [0,1], c_i \in [0,1]$	Severity, confidence
$u_i(t) = \sigma_i e^{-\alpha(t-t_i^{\text{det}})}$	Time-decayed urgency, $\alpha = \ln 2/T_{\text{half}}$
$B_j(t) \in [0, B_{\max}], B_{\max}$	Remaining/max battery
$A_j \in \{0,1\}, \text{status}_j$	Availability (derived from Idle)
$T_{ij}, E_{ij}^{\text{total}}, E_{\text{return}}$	Travel time, total/return energy
$C_{ij}(t), w_T, w_E, w_U$	Assignment cost and weights ($w_T + w_E + w_U = 1$)
$T_{\max}, E_{\max}$	Normalization constants
$x_{ij}(t) \in \{0,1\}$	Assignment variable
$X(t) = (\mathcal{S}, \mathcal{D}, \mathcal{E}, \mathcal{Q})$	Global state

III. System Entities and State Definitions

A. Survivor Task Model

Each survivor detection is modeled as a task $i \in \mathcal{S}(t)$, where $\mathcal{S}(t)$ is the set of all known survivor tasks at time t. Survivor i is defined as:

$$S_i = (x_i, y_i, z_i, t_i^{\text{det}}, c_i, \sigma_i, u_i(t), a_i, \text{assigned\_to}_i) \quad (1)$$

with the following parameters:

Severity σ_i denotes the static medical criticality at detection (independent of time), with $\sigma_i = 1$ representing the most critical condition. Two possible scenarios for obtaining σ_i are: (1) *Scout AI Inference*: estimated automatically from Scout UAV thermal/optical imagery and rubble context via a classification

TABLE II
SURVIVOR TASK PARAMETERS

Parameter	Symbol	Domain
Location	(x_i, y_i, z_i)	$\mathbb{R}^3$
Detection Time	t_i^{det}	$\mathbb{R}^+$
Confidence Score	c_i	$[0, 1]$
Severity	σ_i	$[0, 1]$
Urgency	$u_i(t)$	$[0, 1]$
Assignment Flag	a_i	$\{0, 1\}$
Assigned UAV	assigned_to_i	$\mathcal{D}(t) \cup \{\emptyset\}$

model (e.g., motion/heat signature analysis), and (2) *Human-in-the-Loop*: tagged manually by the Command Center operator reviewing the Scout video feed. If no severity estimate is available, $\sigma_i = 1$ uniformly.

B. Urgency Decay Function

The urgency of survivor i decays exponentially with waiting time according to the Golden Hour principle:

$$u_i(t) = \sigma_i \cdot e^{-\alpha(t - t_i^{\text{det}})} \tag{2}$$

where $\sigma_i \in [0, 1]$ is the severity at detection so that $u_i(t_i^{\text{det}}) = \sigma_i$, $\Delta t = t - t_i^{\text{det}} \geq 0$ is the waiting time, and $\alpha > 0$ is the decay rate. The rate is defined via the half-life T_{half} (e.g., $T_{\text{half}} = 30$ min) as

$$\alpha = \frac{\ln 2}{T_{\text{half}}}, \tag{3}$$

so that $u_i(t_i^{\text{det}} + T_{\text{half}}) = \sigma_i/2$. For $T_{\text{half}} = 30$ min, $\alpha \approx 0.023$ min^{-1} (3.8×10^{-4} s^{-1}); $T_{\text{half}} = 60$ min gives $\alpha \approx 0.0115$ min^{-1}. A survivor found 10 min ago ($\Delta t = 10$) has $u \approx 0.79\sigma_i$, while one found 90 min ago has $u \approx 0.12\sigma_i$ at the same current time t, capturing higher salvageability of fresher detections.

C. Confidence Score and Verification Policy

Each detection carries a confidence score $c_i \in [0, 1]$ representing the Scout UAV detector's estimated probability that the detection is a true survivor (e.g., thermal/optical AI output). A threshold $c_{\text{thresh}} \in [0, 1]$ (e.g., 0.5) separates high- and low-confidence regimes:

$$\text{eligible}(i) = \begin{cases} 1 & \text{if } c_i \geq c_{\text{thresh}} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Detections with $c_i \geq c_{\text{thresh}}$ enter the high-confidence queue $\mathcal{Q}_{\text{high}}(t)$ and are eligible for Responder assignment. Detections with $c_i < c_{\text{thresh}}$ enter the low-confidence queue $\mathcal{Q}_{\text{low}}(t)$ and are *not* eligible for Responder dispatch; they require verification (Scout re-scan or operator review) before promotion to $\mathcal{Q}_{\text{high}}$. In Version 1, $\mathcal{Q}_{\text{low}}$ is modeled as a holding set and verification is abstracted as an exogenous promotion event. The offloading ILP is defined only over $\mathcal{Q}_{\text{high}}(t)$.

D. Responder UAV Model

Each Responder UAV $j \in \mathcal{D}(t)$ is defined as:

$$D_j = (x_j, y_j, z_j, p_j^{\text{base}}, B_j, A_j, s_j, \text{status}_j) \tag{5}$$

with the following parameters:

TABLE III
RESPONDER UAV PARAMETERS

Parameter	Symbol	Domain
Position	(x_j, y_j, z_j)	$\mathbb{R}^3$
Base Station	p_j^{base}	$\mathbb{R}^3$
Remaining Battery	B_j	$\mathbb{R}^+$
Availability	A_j	$\{0, 1\}$
Current Task	s_j	$\mathcal{S}(t) \cup \{\emptyset\}$
Mission Status	status_j	$\{\text{Idle}, \text{EnRoute}, \text{AtSurvivor}, \text{Returning}\}$

E. Availability States

The availability A_j is derived from the UAV's status:

$$A_j = \begin{cases} 1 & \text{if } \text{status}_j = \text{Idle} \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

F. Environment Model

The disaster environment is represented as continuous sets:

$$\mathcal{E} = (V, O) \tag{7}$$

where $V \subset \mathbb{R}^3$ is navigable free space and $O \subset \mathbb{R}^3$ is the static obstacle set (buildings, rubble, no-fly zones), with $V \cap O = \emptyset$. For Version 1, $O(t) = O$ (static) and the hazard map $R : V \to [0, 1]$ is set to $R = 0$ and deferred. A route P is feasible iff $P \subset V$ (all waypoints lie in free space). No discretization or algorithm is assumed at formulation level.

G. Task Queue

The ordered queue of unassigned survivors is defined as a derived view:

$$\mathcal{Q}(t) = \text{order}(\{S_i \in \mathcal{S}(t) \mid a_i = 0\}) \tag{8}$$

ordered by priority:

$$\text{priority}(i) = \beta_1 \cdot u_i(t) + \beta_2 \cdot c_i - \beta_3 \cdot (t - t_i^{\text{det}}) \tag{9}$$

Note: $\mathcal{Q}(t)$ is not an independent state component but a derived view of $\mathcal{S}(t)$ for scheduling. With verification threshold c_{thresh}, the queue partitions as:

$$\mathcal{Q}_{\text{high}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i \geq c_{\text{thresh}}\}, \tag{10}$$

$$\mathcal{Q}_{\text{low}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i < c_{\text{thresh}}\} \tag{11}$$

Only $\mathcal{Q}_{\text{high}}(t)$ is eligible for the offloading assignment; $\mathcal{Q}_{\text{low}}(t)$ is held for verification. *Uncertainty handling:* The partition manages detection uncertainty (false positives from Scout thermal/optical AI). High-confidence detections are directly eligible, while low-confidence detections are buffered in $\mathcal{Q}_{\text{low}}$ for verification (re-scan/operator review) before promotion, preventing scarce Responder UAVs from being wasted on likely false alarms. Note: priority(i) is a pre-filter ordering heuristic for $\mathcal{Q}_{\text{high}}(t)$ (e.g., for greedy fallback or tie-breaking); the optimal assignment is determined by the ILP cost $C_{ij}(t)$, not by priority order.

IV. Decision Variables

A. Assignment Variable

$$x_{ij}(t) \in \{0, 1\} \tag{12}$$

$$x_{ij}(t) = \begin{cases} 1 & \text{if UAV } j \text{ is assigned to survivor } i \text{ at time } t \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

B. Route Variable (Routing Oracle Output)

The route $P_{ij}(t)$ is provided by the Routing Problem as an oracle output, not an offloading decision variable. When queried with $(p_j, p_i, p_j^{\text{base}})$, the oracle returns P_{ij} and $P_{i\to\text{base}(j)}$ if feasible, otherwise infeasibility. For reference, a route is a sequence:

$$P_{ij} = \{p_0, p_1, \ldots, p_K\} \tag{14}$$

where $p_0 = (x_j, y_j, z_j)$, $p_K = (x_i, y_i, z_i)$, $p_k \in V$. The offloading decision is solely $x_{ij}(t)$; T_{ij} and E_{ij} below are consumed from this oracle as $(T_{ij}, E_{ij}) = \text{RoutingOracle}(p_j, p_i)$.

V. Cost Functions

A. Travel Time

$$T_{ij} = T(P_{ij}) = \sum_{k=1}^{K} \frac{\|p_k - p_{k-1}\|}{v_{\max}} \tag{15}$$

where $v_{\max}$ is the maximum UAV speed.

B. Energy Consumption

$$E_{ij} = E(P_{ij}) = \sum_{k=1}^{K} \left(e_{\text{travel}} \cdot \|p_k - p_{k-1}\| + e_{\text{vert}} \cdot \Delta z_k\right) \tag{16}$$

where:

- e_{travel}: Energy per unit horizontal distance
- e_{vert}: Energy penalty for vertical climb
- $\Delta z_k = \max(0, z_k - z_{k-1})$: Vertical gain

C. Service-Related Energy

If the UAV hovers at the survivor location for duration T_{service} (V1 fixed parameter, e.g., $T_{\text{service}} = 300\,\text{s}$):

$$E_{\text{hover}} = e_{\text{hover}} \cdot T_{\text{service}} \tag{17}$$

D. Return Energy

The energy to return from survivor i to the base of UAV j is:

$$\begin{aligned} E_{\text{return}}(i, j) &= E(P_{i\to\text{base}(j)}) \\ &= \sum_{k=1}^{K'} \left(e_{\text{travel}}\|p_k - p_{k-1}\| + e_{\text{vert}}\Delta z_k\right) \end{aligned} \tag{18}$$

where $P_{i\to\text{base}(j)} = \{p_0 = p_i, \ldots, p_{K'} = p_j^{\text{base}}\}$ is the feasible route from p_i to p_j^{base} with $p_k \in V$. This explicitly uses p_j^{base} from the UAV model and is generally *not* equal to $E(P_{ij})$ due to differing start position $p_j \neq p_j^{\text{base}}$ during online operation and asymmetric vertical cost.

E. Total Energy Required

The worst-case (conservative) total energy for a roundtrip mission is:

$$E_{ij}^{\text{total}} = E(P_{ij}) + E_{\text{hover}} + E_{\text{return}}(i, j) \tag{19}$$

where $P_{ij} = P_{j\to i}$ is the outbound route from current position p_j to survivor p_i. Variant for landing at survivor (no return): $E_{ij}^{\text{oneway}} = E(P_{ij}) + E_{\text{hover}}$, deferred to future work; V1 assumes roundtrip.

F. Assignment Cost

The cost of assigning UAV j to survivor i is:

$$C_{ij}(t) = \begin{cases} w_T \dfrac{T_{ij}}{T_{\max}} + w_E \dfrac{E_{ij}^{\text{total}}}{E_{\max}} & \text{if Feasible}(i, j) \\ \quad - w_U u_i(t) c_i & \\ \infty & \text{otherwise} \end{cases} \tag{20}$$

where $w_T, w_E, w_U > 0$ are weighting coefficients (cost weights, distinct from priority weights $\beta_1, \beta_2, \beta_3$; e.g., $w_T + w_E + w_U = 1$ for normalized trade-off), $T_{\max}$ and $E_{\max}$ are normalization constants (maximum feasible time/energy), and the term $-w_U \cdot u_i(t) \cdot c_i$ is the expected urgency reward: higher $u_i \cdot c_i$ reduces cost, incentivizing assignment to more urgent/confident survivors. Infeasible pairs have infinite cost and are excluded from the ILP.

Rationale: The cost is a dimensionless trade-off among (1) response time $w_T T_{ij}/T_{\max}$, (2) energy $w_E E_{ij}^{\text{total}}/E_{\max}$, and (3) expected value $-w_U u_i(t) c_i$ with $u_i(t) = \sigma_i e^{-\alpha(t - t_i^{\text{det}})}$. Normalization makes w_T, w_E, w_U comparable ($w_T + w_E + w_U = 1$ baseline); increasing w_U favors urgent/confident survivors despite distance, while larger w_T, w_E favor nearest, cheapest assignments. ∞ enforces feasibility before optimization.

VI. Feasibility Constraints

A. Battery Constraint

A UAV j can be assigned to survivor i only if it has sufficient energy to complete the entire roundtrip mission plus a safety reserve:

$$E_{ij}^{\text{total}} + E_{\text{reserve}} \leq B_j(t) \tag{21}$$

where $E_{ij}^{\text{total}} = E(P_{j\to i}) + E_{\text{hover}} + E_{\text{return}}(i,j)$ is the roundtrip energy (outbound, hover $T_{\text{service}} = 300$ s, and return to p_j^{base}), $E_{\text{reserve}} = 0.2\, B_{\max}$ is a 20% safety buffer for wind, detours, and model error, $B_j(t) \in [0, B_{\max}]$ is the remaining battery at time t, and $B_{\max}, T_{\text{service}}, e_{\text{travel}}, e_{\text{vert}}, e_{\text{hover}}, v_{\max}$ are global V1 constants. If (21) is violated, Feasible$(i,j) =$ false and $C_{ij}(t) = \infty$.

B. Reachability Constraint

A route P_{ij} exists only if:

$$p_k \in V \quad \forall k \in \{0, 1, \ldots, K\} \tag{22}$$

i.e., the entire path lies in navigable space. Reachability is a boolean oracle provided by the Routing Problem (e.g., 3D A* over $V \setminus O$); offloading consumes exists(P_{ij}) without defining discretization.

C. Availability Constraint

$$x_{ij}(t) \leq A_j \tag{23}$$

D. Feasible Assignment Condition

A complete feasibility check is:

$$\begin{aligned} \text{Feasible}(i,j) = &(A_j = 1) \wedge (c_i \geq c_{\text{thresh}}) \wedge (P_{ij} \text{ exists}) \\ &\wedge (E_{ij}^{\text{total}} + E_{\text{reserve}} \leq B_j(t)) \end{aligned} \tag{24}$$

VII. Assignment Constraints

A. Single Assignment per Survivor

Each survivor in $\mathcal{Q}_{\text{high}}(t)$ is assigned to at most one Responder UAV:

$$\sum_{j \in \mathcal{D}_{\text{avail}}(t)} x_{ij}(t) \leq 1 \quad \forall i \in \mathcal{Q}_{\text{high}}(t) \tag{25}$$

B. Single Task per UAV

Each available UAV is assigned to at most one survivor at a time:

$$\sum_{i \in \mathcal{Q}_{\text{high}}(t)} x_{ij}(t) \leq 1 \quad \forall j \in \mathcal{D}_{\text{avail}}(t) \tag{26}$$

where $\mathcal{D}_{\text{avail}}(t) = \{j \mid A_j = 1\}$ and $N' = |\mathcal{Q}_{\text{high}}(t)|$, $M' = |\mathcal{D}_{\text{avail}}(t)|$. The ≤ 1 allows partial matching: a survivor may remain unassigned if no feasible UAV, and a UAV may stay idle if no high-confidence survivor.

C. Feasibility Enforcement

If a pair (i,j) is not feasible, its cost is set to infinity: $C_{ij}(t) = \infty$ when $\neg$Feasible(i,j). This includes $c_i < c_{\text{thresh}}$, $A_j = 0$, no route, or battery violation. Any optimal solution will then have $x_{ij} = 0$ for such pairs. No separate $x_{ij} \leq A_j$ is needed.

VIII. Objective Function

A. Primary Objective: Minimize Cost

The snapshot offloading problem OFF-SNAP(t) over $\mathcal{Q}_{\text{high}}(t) \times \mathcal{D}_{\text{avail}}(t)$ is:

$$\min_{\mathbf{x}(t)} \sum_{i \in \mathcal{Q}_{\text{high}}(t)} \sum_{j \in \mathcal{D}_{\text{avail}}(t)} x_{ij}(t) \cdot C_{ij}(t) \tag{27}$$

If $\mathcal{Q}_{\text{high}}(t) = \emptyset$ or all $C_{ij} = \infty$, the optimum is $x_{ij} = 0$ (no dispatch, wait for next event).

IX. Scope and Limitations

A. Core Model Includes

- Survivor detection as exogenous input
- Responder UAV assignment and routing
- Battery-aware feasibility
- Time-dependent urgency
- Online, event-triggered decision making

B. Excluded (Future Work)

- Dynamic aftershock obstacles
- UAV-to-UAV collision avoidance
- Communication relay UAVs
- Hot-swap battery stations
- Detection uncertainty verification policies
- Recourse/reassignment after dispatch

X. Conclusion

This document presents a complete mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM) for post-earthquake UAV-assisted response. The model captures the essential dynamics of the system: survivor states with time-dependent urgency, Responder UAV states with battery constraints, and the online nature of detection arrivals and resource availability. The formulation separates feasibility (battery and routing constraints) from optimization (assignment and path selection), providing a clear foundation for algorithm design and simulation. The coupling between assignment and routing is explicitly defined through the cost function C_{ij}, which depends on both travel time and energy consumption.